

JULIUS CAESAR F. RAAGAS

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<https://julius-raagas.vercel.app/>

Technical Skills

- **AI Engineering:** OpenAI API, LLM Integration, RAG, Embeddings, Semantic Search, Supabase pgvector, LangChain, Prompt Engineering, AI-Assisted Development Workflows, AI Output Evaluation
- **Frontend Development:** JavaScript, TypeScript, React.js, Next.js, Vue.js, Nuxt.js, Quasar Framework, Tailwind CSS, SCSS
- **Backend Development:** Node.js, Express.js, Laravel, PHP, FastAPI, RESTful API Development, Authentication, API Integration
- **Databases & Vector Search:** PostgreSQL, MySQL, MongoDB, Firebase, Supabase, Prisma ORM, Pinecone
- **Cloud & DevOps:** AWS S3, AWS EC2, Vercel, Firebase Hosting, Docker, Docker Compose, GitHub Actions, CI/CD Pipelines
- **Testing & Tools:** Git, GitHub, Postman, Playwright, Jest, Figma, WordPress, XAMPP, Technical Documentation, Github Actions, Cursor, Codex, Claude

EXPERIENCE

Pixel8 Web Solutions & Consultancy Inc.

Aug 2023 - Dec 2023

Software Engineer Intern

Legazpi City, Albay, Philippines

- **Frontend Development:** Developed and enhanced multiple UI features using the Quasar Framework (Vue.js), focusing on improving user interaction, responsiveness, and component reusability. Collaborated with the design team to translate wireframes into dynamic and maintainable frontend modules that followed modern UX principles. Built and integrated RESTful APIs (PHP/XAMPP) to handle dynamic content and data synchronization between client and server.
- **Backend & API Management:** Designed, implemented, and tested RESTful APIs using PHP/XAMPP, ensuring reliable communication between frontend interfaces and backend data services. Integrated database queries, validated endpoints, and documented routes for consistent team use.
- **Integration & Testing:** Conducted thorough integration testing to ensure seamless data flow and eliminate compatibility issues between frontend and backend systems. Utilized version control and debugging tools to identify and resolve performance bottlenecks.
- **Leadership & Mentorship:** Supervised a small team of OJT interns, guiding them in following best coding practices, debugging issues, and documenting their work. Provided peer reviews and led short discussions on improving workflow efficiency.
- **Documentation & Knowledge Transfer:** Created detailed documentation for UI components, API endpoints, and workflow processes to streamline onboarding and ensure long-term maintainability for future developers.

Macrologic Diversified Technologies Inc.

Aug 2024 - Dec 2024

Web Developer Intern

Bacoor, Calabarzon, Philippines

- **Website Management & Enhancement:** Managed, maintained, and enhanced the company's WordPress website, implementing structured UI improvements and optimizing modules based on user behavior and team feedback. Collaborated on layout redesigns to make the site more intuitive, accessible, and visually appealing.
- **Performance Optimization:** Integrated lazy loading, optimized media assets, and fine-tuned caching mechanisms, which significantly improved load times and overall site responsiveness. Conducted browser compatibility tests and implemented SEO-friendly practices to enhance visibility and speed metrics.
- **Cross-Team Collaboration:** Worked closely with marketing and design teams to align content updates with campaign timelines and branding consistency, ensuring that each update met both aesthetic and functional objectives.
- **Documentation for Future Development:** Authored comprehensive documentation covering plugin usage, theme configurations, and page structures to assist future developers and interns. This ensured a smooth handover and maintained consistency across future updates.

- **Full-Stack Web Development:** Designed and developed end-to-end web applications for school, research, and small business projects, building responsive frontends, backend logic, secure APIs, database workflows, and deployment-ready solutions using PHP, JavaScript, Node.js, Express.js, and related web technologies.
- **AI Integration & Engineering:** Integrated AI-powered features into web applications, including LLM-based workflows, prompt-driven automation, text processing, semantic matching, and AI-assisted content generation. Worked with AI APIs and custom logic to connect model outputs with real application features.
- **UI/UX Design:** Created intuitive, mobile-friendly interfaces emphasizing accessibility, clarity, and visual appeal. Used **Figma** and **Tailwind CSS** to prototype and implement consistent UI patterns aligned with modern design standards.
- **Model Training & Data Processing:** Assisted in preparing datasets, cleaning and preprocessing data, testing model behavior, and improving AI outputs through structured prompts, evaluation, and iteration. Applied basic model training and fine-tuning concepts for academic and prototype-level AI systems.
- **UI/UX Design:** Created intuitive, mobile-friendly interfaces with a focus on accessibility, clarity, and usability. Used Figma and Tailwind CSS to prototype layouts and implement consistent UI patterns aligned with modern design standards.
- **Client Collaboration & Documentation:** Worked directly with students, clients, and small institutions to gather requirements, present mockups, refine features, and deliver technical documentation covering setup, deployment, maintenance, and handover processes.

PROJECTS

Heartopia Draft | Full-Stack Developer / Sole Developer

2025

<https://heartopia-draft.vercel.app/>

- Designed and built a production-grade 2D plot and house-layout planner using Next.js 16, React 19, TypeScript, Tailwind CSS, HTML5 Canvas, MongoDB, JWT authentication, and Jest.
- Developed a custom canvas editor with pan, zoom, touch gestures, precision grid snapping, wall/floor placement, doors, windows, fences, furniture, crop tiles, undo/redo, and autosave.
- Engineered a metric-first coordinate system where layout logic is handled in real-world units instead of pixels, improving accuracy across zoom levels and screen sizes.
- Built advanced spatial logic including shared-edge wall handling, opening safety-margin validation, furniture collision detection, 8-direction rotation, single drag, and marquee group selection.
- Implemented full-stack features such as REST API route handlers, MongoDB persistence, authenticated cloud saving, demo mode, draft migration on sign-up, rate limiting, validation, and protected write paths.
- Maintained a clean layered architecture with separated domain logic, reusable hooks, and circular-dependency checks using madge to keep the codebase scalable.
- Created and maintained a strong test suite with 860+ test cases across 153 test suites, covering spatial math, React hooks, API handlers, validation, rate limiting, autosave, and UI behavior.
- Demonstrated end-to-end ownership across product design, system architecture, frontend engineering, backend development, security, testing, and deployment readiness.

Stillness E-commerce | AI-Native Full-Stack Developer / Sole Developer

2024

<https://stillness-ecommerce.vercel.app/>

- **Rapid Development Milestone:** Engineered and deployed a fully functional, minimalist e-commerce platform in a record 72-hour timeframe by combining years of developer experience with AI-augmented engineering.
- **AI-Driven Workflow:** Leveraged advanced Prompt Engineering to automate component scaffolding, architect complex Vuex/Pinia state management, and generate optimized SCSS/CSS layouts, effectively reducing development time by 80%.
- **Frontend Excellence:** Built with Vue.js, focusing on high-performance reactivity and a seamless mobile-first shopping experience; deployed via Vercel for optimized global delivery.
- **Core Functionality:** Developed a dynamic product catalog, an integrated shopping cart system, and a clean, conversion-focused UI/UX using modern design principles.
- **Strategic Execution:** Demonstrated the ability to bridge the gap between architectural logic and AI execution, ensuring production-ready code quality while maintaining an ultra-fast delivery pace.

- **Project Scope:** Co-developed a community-driven social media platform for culinary enthusiasts, enabling users to discover, share, and interact with regional recipes.
- **Frontend Engineering:** Architected a high-performance, responsive user interface using **Vue.js** and the **Quasar Framework**, ensuring a seamless cross-platform experience (Web/Mobile) through custom component libraries.
- **System Architecture:** Developed the complete backend infrastructure using **Node.js** and **Express.js**, managing end-to-end data flow and secure **RESTful API** development.
- **Core Feature Engineering:** Built a comprehensive social ecosystem from scratch, including secure user authentication, interactive commenting, and a personalized "bookmarking" system.
- **Integration & Synchronization:** Spearheaded the integration of backend services with the frontend, ensuring real-time data synchronization for user profile management and media uploads.

CERTIFICATES

Pixel8 Web Solutions & Consultancy Inc.

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| • Frontend Development Certificate | Dec 2023 |
| • Backend Development Certificate | Jan 2024 |
| • Professional Development Certificate in Software Engineering | |